AI INNOVATION CHALLENGE MARCH 2026

ClearStep

AI that turns confusing information into calm, actionable steps

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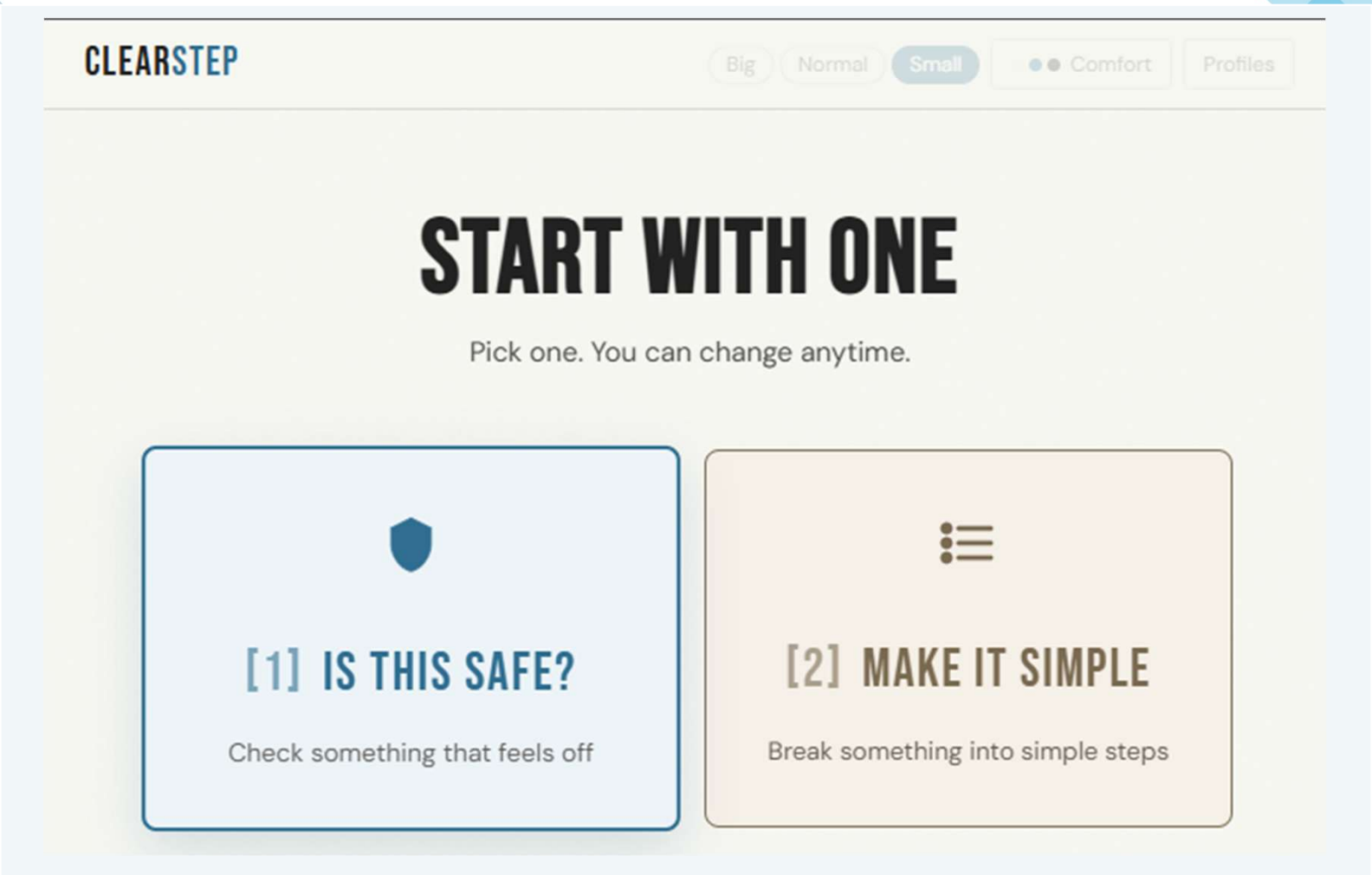

CLEARSTEP
Big
Normal
Small
Comfort
Profiles
START WITH ONE
Pick one. You can change anytime.
[1] IS THIS SAFE?
Check something that feels off
[2] MAKE IT SIMPLE
Break something into simple steps

THE PROBLEM

1 **Confusing, high-pressure information**

Users receive messages that feel urgent and unclear.

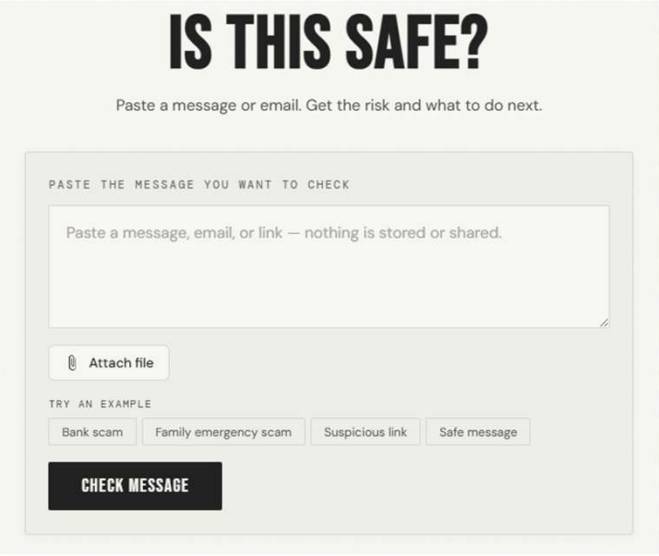
2 **Detection without guidance**

Existing tools label risk but do not tell users what to do next.

3 **Decision burden falls on the user**

People must interpret, decide, and act, often under stress.

Suspicious messages



Complex instructions

THE SOLUTION

Action First AI

MODE 1 — IS THIS SAFE?

- Detects risk level
- Explains why it is risky
- Tells you exactly what to do

HIGH RISK

Fake bank message trying to steal your login information.

KEY SIGNALS

- Bank impersonation
- Urgent language
- Suspicious link

NEXT STEPS

1. **Do not click the link**
2. Call Chase directly using official number

Why this result?

- "Bank impersonation" was identified in this message.
- Uses words like "immediately" to stop you thinking.
- Contains a link that may not go where it says.
- ClearStep is an AI tool. Always verify with a professional if unsure.

MODE 2 — MAKE IT SIMPLE

- Breaks down complex information
- Separates warnings from actions
- Guides step-by-step

CLEAR

Benefits enrollment deadline requires completing five steps to maintain health coverage.

2 of 5

Select health insurance plan

Reminder + Add reminder

✓ MARK AS DONE

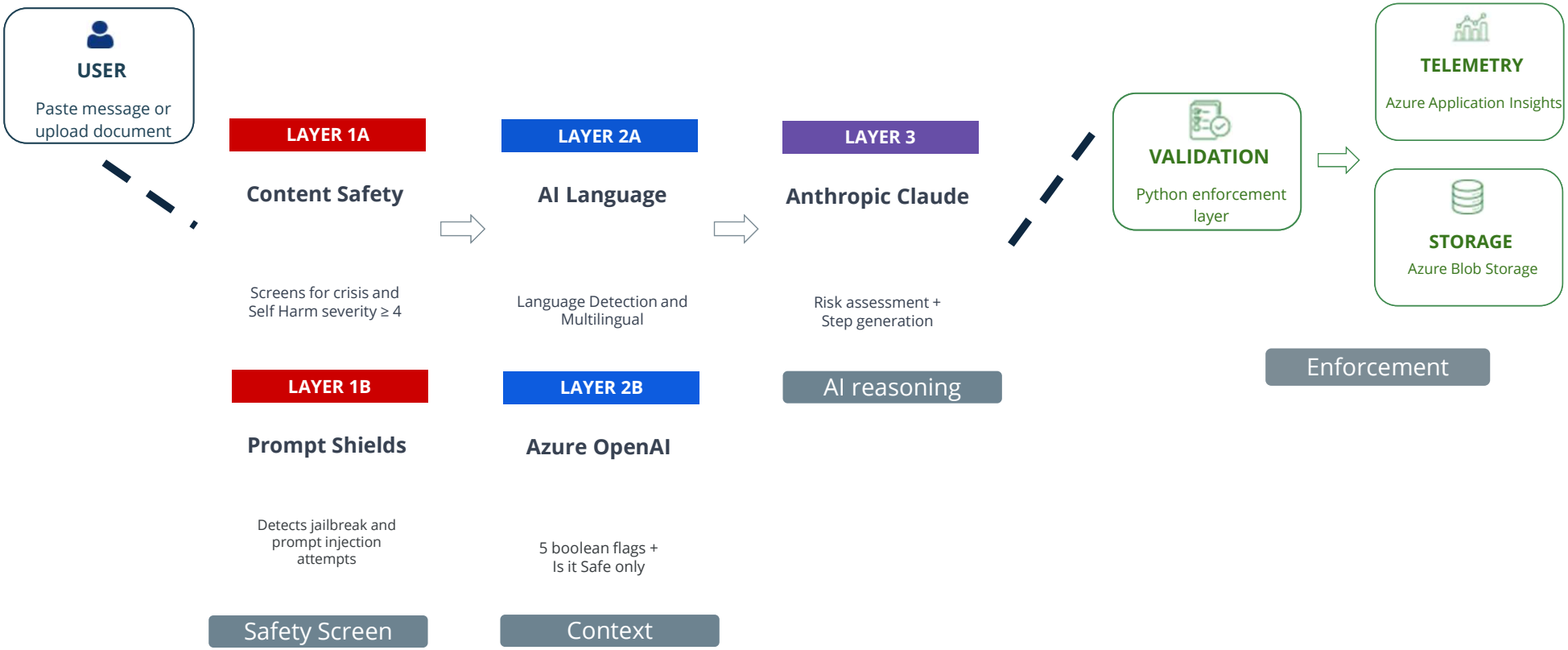
✓ Step 1 done Undo

1. ~~Log into HR portal~~ Undo
2. **Select health insurance plan** Now
3. Upload dependent verification documents
4. Confirm payroll deductions
5. Submit enrollment

Why this result?

3-LAYER AI PIPELINE

Every request passes through a strict 3-layer sequence. Robust error handling ensures no system crash



POWERED BY AZURE AI & CLOUD SERVICES

Built using 10 Azure services across safety, intelligence, and infrastructure

Azure OpenAI
Signal classification, 5 Boolean flags

Content Safety
Crisis screening + jailbreak detection before LLM

Blob Storage
Audit log and no raw content

Azure AI Language
Multilingual detection

App Service
Managed hosting + CI/CD

Azure Key Vault
Secret Management, no keys in code

Azure AI Speech
Text-to-speech

Azure Application Insights
28 custom telemetry events

Azure Cosmos DB
Accessibility preferences

Foundry
Model deployment platform

ALL AI PRINCIPLES ARE INTEGRATED

Fairness	5 accessibility palettes for specific neurological needs. Varying Reading level. Multi Language detection.
Reliability & Safety	Crisis and medical responses are enforced in Python. Strict schema validation and XSS output sanitization. Automated fallback and rate-limiting to prevent abuse
Privacy & Security	No message content stored. Anonymous session ID. No accounts, no tracking. No PII.
Inclusiveness	Plain language. Built for various users (elderly, neurodivergent, non-English speakers).
Transparency	'Why this result?' panel for every output. Persistent AI disclaimer and fallback indicator unavailable Ai services. Automated deference to the original document for medical content.
Accountability	All actions are auditable. Every analysis, flag, and AI decision is logged to Azure Blob Storage for traceability and responsible review.
Human Oversight	Medical and legal content defers to real professional. Features hardcoded crisis routing (988). User controls every transition, step engine never auto- advance.

ACCESSIBLE DESIGN IS THE PRODUCT

Five neuro-inclusive palettes engineered for cognitive ease and covering safe, caution, danger and medical states

Calm Default
General users
No pure white

Low Sensory
Autism-friendly
Zero red or orange

Dyslexia
Dyslexic readers
Cream, reduced glare

High Focus
ADHD
Single accent only

Dark Mode
Photosensitivity
Navy, not pure black

Adaptive Reading Level

A tri-level toggle (Small/Normal/Large) that dynamically adjusts font size, line height, and AI output density. Preferences persist across sessions via Cosmos DB

MEDICAL

Medication
duration ins

Medical information
This is a reminder tool only. Alwa
pharmacist or doctor with any qu

BEFORE YOU START

→ 7-day course

START WITH STEP 1 →

MEDICAL

Medication schedule with specific tir
duration instructions.

Medical information
This is a reminder tool only. Always follow your original prescription a
pharmacist or doctor with any questions.

BEFORE YOU START

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OUR APPROACH TO THE PROBLEM

1 Start with the person not the problem

What does it feel like to receive a frightening message when your brain processes anxiety differently?

2 Ask about experience, not diagnosis

UI adapts to how you process information, not how you are diagnosed

3 Calm language is a technical requirement

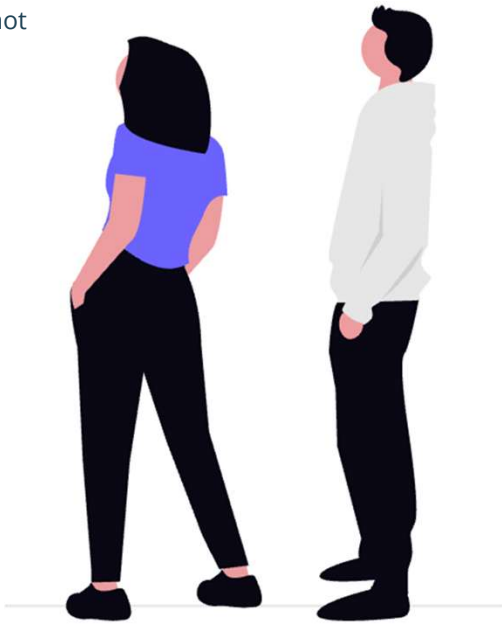
Tone of AI output is treated as a critical component of the design

4 Explainability reduces anxiety and create trust.

Showing the “why” behind every model output to empower the user

5 Design for Failure and Safety

Graceful error states, fallback flows, and transparent limitations, so users stay informed, not abandoned.



KEY LEARNINGS

Labeling without action is Incomplete	A "High Risk" label without a next step increases anxiety. ClearStep provides "Action-First" steps to eliminate decision paralysis
Accessibility First is Universal Design	Solving for cognitive overload by designing a simple, intuitive experience that empowers neurodivergent users and reduces complexity.
Fail-Safe Architecture	We prioritize uptime, security and system integrity by combining 'Zero-Secrets' architecture via Azure Key Vault, protecting user data with a layered Fallback (Azure → Anthropic → Local Scorer).
Ethics as a Pipeline	Integrated all six Responsible AI Principles (Fairness, Reliability, Privacy, Inclusiveness, Transparency, Accountability) directly into the engineering workflow.

ROADMAP

PHASE 1: HIGH PRIORITY

Azure Computer Vision

Finalizing OCR endpoint configuration and schema validation for vision Integration.

PHASE 2: PLANNED

Session History

Optional anonymous history log for users to revisit past analyses without account creation or tracking.

PHASE 3: FUTURE

Browser Extension

Integrate "Is This Safe?" tool to any email client, allowing users to verify suspicious messages without leaving their inbox.

ClearStep

Reducing cognitive load for neurodivergent users

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Research Support, Feature ideation, UX concept exploration

Live Demo
Github Repository